Chapter 6

Conclusions and Outlook

The thesis asked whether the volatility of an organic polymer-electrolyte memristive device can be controlled and put to use. Chapters 2 to 4 establish the device response, diagnose its reproducibility, and measure how composition sets its relaxation time. Chapter 5 tests what that time constant is good for, first as a designed filter element and then as a node of a reservoir, each against linear references. The conclusions below separate the measured device evidence from the model-based results and identify the experiments needed for a hardware test.

6.1 Summary of Contributions

The experimental and computational contributions differ in evidentiary strength because only the composition grid is replicated across devices.

A characterised synaptic platform (chapter 2). A reversible two-terminal organic memristive platform combines *Super Yellow*, *Hybrane*, and LiOTf in a solution-processed active layer between ITO and Ag electrodes. Across assay-specific junctions sharing that formulation and

architecture, it supports analogue I–V hysteresis (an 85 % conductance increase from the first to the second sweep at 1 V, growing ∼15-fold over ten cycles), reversible potentiation and depression exceeding 200 %, and two distinguishable retention regimes. Recalculation of the archived positive-pulse train gives 0.084 µJ–1.55 µJ per event as the state changes (median 1.08 µJ). Standard-exponential fits reported in the published SI give $\tau_S \approx 2.5$ s–3 s for short-term memory and $\tau_L \approx 4.7$ s for long-term memory. The formulation also supports one numerically separated four-state excitatory post-synaptic current sequence and an asymmetric Hebbian STDP window; direct re-fitting of the five-junction mean gives branch time constants of 89 ms and 150 ms. A fast-triflate/slow-lithium account is a plausible candidate interpretation, but ion motion was not measured and the later cation survey does not corroborate its simple ordering. The assays were independently peer reviewed [1], but they do not constitute a population study; table A.1 gives the specimen mapping.

A reproducibility diagnosis and a fabrication-record methodology (chapter 3). Successive batches of nominally identical *Hybrane* devices changed from low-current, high-area, multilevel behaviour to a high-conductivity response without a usable switching window. Per-device weighting, restriction to a standard protocol, sweep-amplitude matching, and recovery-tail sensitivity confirmed a multi-feature change: the on–off ratio fell from 2.44 to 1.21, normalised hysteresis area from 0.235 to 0.027, and potentiation at 3 V changed from +17.9 % to −2.1 %. Conduction also became more ohmic at a matched 1.2 V. The change followed fabrication date rather than device age, because finished *Hybrane* devices retained their characteristics over weeks. The evidence points to ageing of the *Hybrane* stock, with *Super Yellow* excluded, under a moisture-driven hydrolysis hypothesis supported by an annealing-recovery test. Replacing *Hybrane* with poly(ethylene oxide) increased the normalised window area from 0.26 to 0.42 at 3 V and maintained usable switching across roughly

two and a half years. The study also produced the fabrication-record standard, normalised characterisation procedure, automated feature pipeline, and trend-analysis tools used in the later chapters.

Compositional and chemical control of the dynamics (chapter 4). The replicated *Super Yellow*/PEO/LiOTf/Ag grid shows that composition tunes fading-memory time over roughly 2 s–20 s; higher ion-transport-polymer content gives a weaker, faster-relaxing response. Host (PEO versus TMPE) and anion (triflate versus TFSI) extend the range in the illustrative chemistry survey. Cation identity also shifts the dynamics, but the triflate series runs opposite to the hypothesised $Li^+ > Na^+ > K^+$ order and changes again with the anion. The available data therefore do not establish a transferable cation law. Drive amplitude changes the apparent timescale, so quantitative comparisons require a matched protocol. Equilibrium impedance falls along the composition axis and tracks fading-memory time, and its low-frequency capacitance exceeds the geometric capacitance by a median factor of about 69, consistent with seconds-scale relaxation arising from ionic storage within the film. ATR, XRD, and UV-Vis associate the host effect with short-range ether order in an X-ray-amorphous, salt-dissolved film without a shift in the *Super Yellow* optical gap. Because the PEO fraction changes both fading memory and input nonlinearity, each composition defines a paired operating point. Read as circuit elements, the replicated cells give two to five time constants that can be selected by formulation and told apart despite a within-cell tolerance of $\sigma(\ln t_{1/2}) \approx 0.26$.

Component cards and two uses of the time constant (chapter 5). Each replicated cell was summarised as a component card: half-life, stretch, the *RC* time constant with the same half-life, and the corner frequency of its measured kernel. The cards span time constants of about 4

to 28 s and corners of 0.002 Hz–0.04 Hz; stretched cells have lower corners than their half-life suggests. Compact behavioural models ($\phi \otimes \lambda \otimes f$) that follow the full measured relaxation were then evaluated in simulation in two roles. As a designed filter, the fastest cell removes the tonic level of electrodermal signals as well as a digital filter (synthetic SCR detection F1 0.713 against 0.701; WESAD chest stress-versus-baseline AUC 0.952 against 0.951) and behaves as an RC element with the same half-life; the measured stretch never improves on that element, and two-composition band-passes fall below the phasic band. As reservoir banks, mixed compositions raise random-input linear memory from 3.22 to 4.44 and information-processing capacity from 5.18 to 9.77 (all ten seeds), and add 0.023 R^2 on multi-lag WESAD reconstruction over the best single-composition bank. A linear exponential-moving-average (EMA) bank at the same half-lives does better on that reconstruction (0.742 against 0.707) and matches the device banks under injected noise. The device banks exceed the linear controls on the sustained three-class labels (0.562 against 0.508), where they remain well below a 2018 engineered-feature pipeline. Seeds quantify algorithmic sensitivity rather than experimental replication, and all task results are simulations of measured device models.

6.2 Answers to the Thesis Questions

Realising and understanding an exemplar. *Can a reversible two-terminal organic memristive platform made from a solution-processed polymer-electrolyte composite support analogue switching, potentiation and depression, distinct retention regimes, EPSC, and STDP responses, and can cation–anion mobility asymmetry account for the retention regimes?* The common formulation and architecture of chapter 2, fabricated by solution processing and thermal evaporation, support these responses across assay-specific junctions. Differential displacement and relaxation of tri-

flate and lithium are consistent with the two regimes but are not identified by the electrical measurements. The reverse and anion-dependent cation ordering in chapter 4 means that the proposed simple mobility hierarchy is not corroborated; species-selective measurements remain necessary.

Composition as the quantitative control. *How do the dynamics depend on composition, and does moving the cation along the Li^+, Na^+, K^+ series shift the timescale as coordination chemistry suggests?* The replicated PEO/LiOTf grid shows a monotonic composition dependence: the ratio tunes fading-memory time over roughly an order of magnitude, with switching window and potentiation strength moving in the same direction. Cation identity shifts the dynamics, but the triflate retention series runs opposite to the predicted $Li^+ > Na^+ > K^+$ order and changes again with the anion. Sample size and protocol/electrode confounds prevent a transferable cation law. Coordination chemistry remains a qualitative interpretation rather than a quantitative result.

Utility for temporal processing. *What can the composition-set time constant be used for, how does a device compare with an RC element of the same half-life, and does a bank of mixed compositions compute anything that linear filtering does not?* In the simulations of chapter 5, the time constant is usable as a component value: a cell chosen by its corner frequency performs baseline removal on electrodermal signals at the accuracy of a digital filter and is equivalent to an RC element of the same half-life. Mixed-composition banks broaden random-input memory and nonlinear capacity, but on the physiological tasks a linear filter bank at the same time constants matches or exceeds them except on the sustained three-class labels. The time constants themselves carry most of the value.

6.3 Benchmarking Against the Right Reference Class

Because the device state relaxes over seconds, two reference classes apply: conventional filter components, for the designed-filter role, and temporal and event-driven neuromorphic systems, for the reservoir role [2–4]. Non-volatile DRAM, SRAM, and crossbar ReRAM serve different storage functions. Current physical-reservoir comparisons must also separate the physical node, input encoding, output layer, and network architecture [5]; task scores from different layers and protocols are not a common leaderboard.

Against conventional filters. As a filter, a composition-selected device performs like an RC element of the same half-life and like a digital filter, so it offers no accuracy advantage. Its possible advantage lies in how the time constant is obtained: a $0.0825\,\mathrm{cm}^2$ two-terminal film reaches seconds with no separate capacitor, whereas time constants of hundreds of milliseconds already require large capacitors or very small currents in analogue CMOS [6], and printed circuits have relied on printed low-pass filters for temporal processing, including printed electrodermal stress detection [7, 8]. Against these references the device has measured costs: write voltages of 3 V–4 V in the PEO protocols, a write energy of about $1\,\mu\mathrm{J}$ per event in the proof-of-concept junction, a within-cell tolerance of $\sigma(\ln t_{1/2}) \approx 0.26$, and unencapsulated stability measured over hundreds of hours.

Against physical reservoirs. Physical reservoir computing has been demonstrated in dynamic memristors [9, 10], organic electrochemical devices [11], and delay-coupled single nodes [12]. More recent systems add direct electrical tuning in two-terminal oxide devices [13], gate- and

wavelength-controlled organic transistor dynamics [14], and an organic electrochemical reservoir with a physical device read-out [15]. These systems require fading memory and nonlinearity [4, 16]. The present devices have measured fading memory, while the fitted models have nonzero nonlinear information-processing capacity. Their sub-kilohertz response overlaps many physiological and environmental signals without time rescaling. Composition changes the fading-memory time over 2 s–20 s; in the simulations, the heterogeneous model bank raises linear and nonlinear capacity, but linear filters at the same time constants match it on most physiological tasks.

Against organic neuromorphic devices. Much of organic neuromorphic electronics targets linear, symmetric, non-volatile conductance updates for weight storage, as in the ENODe family [17–19]. The devices studied here use ion-relaxation timescales for temporal processing and are compared with ionic and electrolyte-gated organic neuromorphic systems [11, 20].

Scope of the claims. The experimentally supported claim is that composition sets seconds-scale fading memory in this two-terminal organic platform, with a measured tolerance. The computational claims are that the measured relaxation, used as a designed filter, performs slow-signal baseline removal as well as a conventional filter, and that banks of behavioural models add little to linear filtering at the same time constants on the physiological tasks studied. The application scores are in silico and do not establish the performance of a wired circuit; no priority is claimed for a first tunable two-terminal reservoir, and no affective-computing advantage is claimed. The simulated reservoir contains independent, non-recurrent nodes; its absolute NARMA errors are high and its capacities remain below the node-count bound. A fabricated device filtering a

Table 6.1: Recent directly relevant reference points for tunable physical reservoirs. Temporal ranges are the values reported under each study's own extraction protocol; tasks, read-outs, and metrics differ, so the table calibrates scope rather than ranking performance. "Physical" means that measured device states formed the reservoir; it does not imply that every input/output operation was analogue or on-chip.

Study	Node and terminals	Tuning handle	Reported temporal range	Reservoir evidence and task scope
This thesis	*Super Yellow*/PEO/LiOTf organic composite; two-terminal	PEO/salt composition; drive amplitude	2 s–20 s across the replicated composition grid	Single-node dynamics measured; designed filters and reservoir banks simulated on WESAD/PhysioNet tasks against linear controls; no wired circuit
Gao et al. (2024) [14]	Organic vertical field-effect transistor; three-terminal optoelectronic	Input wavelength and gate bias	0.005 s–13.2 s with optical input and gate bias	Physical grouped-reservoir states; image recognition and time-series prediction
Han et al. (2025) [15]	Solid-electrolyte OECT; three-terminal	Ethylene-glycol content selects volatile or non-volatile operation	No common fitted-τ range reported	Physical reservoir and physical device read-out; three-class human-activity classification
Lee et al. (2025) [13]	Pt/HfO_2/TiN memristor; two-terminal	Inter-pulse base voltage	Fitted τ: 0.92 ms to 123 ms; inputs 0.14 ms–100 ms	Physical reservoir states; N-MNIST classification and Mackey–Glass prediction

recorded physiological signal remains the missing end-to-end test.

6.4 Limitations of the Evidence Base

The conclusions are limited by the evidence available for each part of the thesis.

1. **Extended protocols are proof-of-concept.** EPSC, separated short-term memory and long-term memory retention, and STDP exist only for the chapter 2 *Hybrane* formulation; some specimen identifiers are not resolved in the recovered archive, and the available multi-junction data share one substrate. They function as formulation-level lithium priors and checks in the later chapters and were not extended to population-level comparison across the PEO family.

2. **Only composition is replicated.** The quantitative core is the *Super Yellow*/PEO/LiOTf/Ag composition grid. The host, anion, and cation comparisons rest on one or two screened devices per matched cell and are illustrative. The electrode contrast is also illustrative and partly confounded with fabrication generation; its negative-polarity measurements are single-composition and uncurated, and its frequency-domain measurements rest on one batch with substrate-confounded cations. No powered material law, including a $Li^+ > Na^+ > K^+$ ordering, is claimed.

3. **Protocol-specific timescales.** Because the drive amplitude cosets the apparent fading-memory time, absolute retention values are protocol-specific, and the supra-threshold read used in the comparative corpus differs from the sub-threshold read of the proof-of-concept device.

4. **The composition assumption behind the models.** Pulse integration and state relaxation were characterised separately, at different write amplitudes (potentiation at 3 V, decay at 4 V) and one inter-pulse cadence. Combining them in the behavioural models assumes that the two regimes are compatible. The nominal composition cards already have an across-card write–retention association stronger than the central estimate transferred from the gold/TMPE corpus ($\rho = +0.46$), so that estimate maps to $\kappa^\star = 0$ in the sensitivity model. Across the tested coupling range, the heterogeneous-to-homogeneous linear-capacity ratio stays between 1.29 and 1.42. The designed-filter results further assume that the measured large-signal relaxation also governs small signals around a written state. A matched protocol and a small-signal relaxation measurement are still needed for the lead composition.

5. **The application results are in silico and only partly cross-corpus.** Every temporal-processing result is a simulation of measured device models with a linear read-out; no wired filter or reservoir was built. The physiological results use WESAD as the main corpus, with synthetic ground truth for the filters and the PhysioNet Non-EEG corpus for noise robustness. Linear filters at the same time constants match or exceed the device banks on most of these tasks, and the device models remain below an engineered-feature pipeline on the WESAD labels.

6. **Raw-data availability is incomplete.** The repository verifies the committed analysis code, derived tables and figures, and complete thesis build, but the experimental archive is separately held and unversioned. A SHA-256 manifest binds the principal processed database snapshot without covering every raw measurement. WESAD and PhysioNet Non-EEG must also be obtained independently; CASE is excluded because its raw corpus was unavailable and no

CASE result is reported. The exact boundary and regeneration commands are recorded in section D.3.

7. **Material stability remains open.** Devices were characterised unencapsulated on a ∼300 h ambient shelf scale; cycling endurance, read-disturb under continuous pulsing, and long-run drift under encapsulation were not systematically studied, and device ageing was not modelled in the comparative corpus.

6.5 Outlook and Future Directions

The immediate experimental work follows from the filter role identified in chapter 5. The model assumptions, the chemistry comparisons, stability, and integration follow.

Filter a recorded signal with a fabricated device. Recorded electrodermal signals, converted to the write protocol of chapter 4, could be replayed through a lead-row or fast-row device and read sub-threshold. Comparing the measured output with the simulated filter and with a digital reference on the same recordings would test the designed-filter result end to end.

Measure the small-signal relaxation. A small write superposed on a written state, at several operating points, would show whether small signals relax with the measured large-signal kernel or with the faster kernel predicted by the composed model. The answer fixes the corner frequency of a device used as a filter and decides whether the fastest cells already reach the sympathetic band.

Compare with a printed *RC* element. A device and a printed *RC* element realising the same time constant on the same substrate would settle the practical question: footprint, drive voltage, energy per update, tolerance, and drift over time.

Matched pulse protocol. A matched-amplitude pulse train should vary pulse count and inter-pulse interval together for one composition, electrode, and read protocol. The gold/TMPE negative-polarity corpus of chapter 4 already records integration and relaxation within single runs and finds interval-dependent potentiation and drive-depth-dependent relaxation. Repeating this measurement on the lead silver composition with a sub-threshold read would supply the missing joint parameters for the behavioural model.

Replicated chemistry survey. A cation series in one host and anion, at fixed protocol and electrode, with a sub-threshold read and at least three replicates per cell, would test the $Li^+ > Na^+ > K^+$ hypothesis at adequate power. The silver TMPE retention follows the opposite order, K>Na>Li; the gold steady-state window is flat; and the gold negative-polarity data resolve no order. The lock-in corpus contains one cation-ordered candidate: a $\approx 1.4\times$ K>Na>Li junction-capacitance trend at one substrate per salt (chapter 4). Profilometry of the existing substrates and a replicated series could test whether this trend is chemical or geometric. Chemistry- and bias-resolved impedance and a generation-controlled silver-versus-gold split would strengthen the microscopic and electrode comparisons.

Test the degradation chemistry. The moisture-driven hydrolytic-aging hypothesis for the *Hybrane* stock (chapter 3) cannot be tested directly because the aged consumable no longer exists in its degraded state. FTIR, GPC, Karl-Fischer/TGA, and EIS on retained aged-versus-

fresh stock would test the hypothesis in future material campaigns. The fabrication-record methodology now records reagent identity and history.

Extend the range of time constants. The replicated grid offers corners below about 0.04 Hz. The faster host and anion variants of chapter 4 (TMPE near 4 s, TFSI near 1 s) point to elements fast enough to set the upper edge of the sympathetic band, and modified polyether hosts could extend memory toward the minutes-scale tonic band. Replicated series would be needed to turn these survey values into component values, and to separate memory time from the input nonlinearity that composition currently changes with it.

Stabilise for long-run inference. Encapsulation and systematic measurements of ageing, endurance, and read disturb are required to assess continuous operation and remove uncontrolled ageing from later comparisons.

Build a reservoir only where linear filtering falls short. A wired bank of composition-diverse devices with an on-chip linear read-out would test the reservoir simulations directly. The simulations suggest that such a bank is worth building for tasks like the sustained WESAD labels, where device banks exceeded linear filters, and not for reconstruction or noise suppression, where a linear filter bank at the same time constants was as good or better.

Physiological sensor interfaces. Solution processing, mechanical flexibility, biocompatible chemistry, and seconds-scale dynamics make these devices candidates for skin-conformal sensing and inference. In the frequency-domain measurements of chapter 4, a DC-biased junction acts as a voltage-programmable low-pass amplifier below 10 Hz. A hardware

experiment could test whether one junction can combine pre-amplification and temporal filtering at a physiological sensor interface.

6.6 Closing Remarks

The replicated experiments identify composition as a control of seconds-scale fading memory in these polymer-electrolyte composites; the cation trend expected from coordination chemistry is not transferable across the available data. Read as component values, the compositions give a printable two-terminal element whose time constant is chosen at formulation. In simulation, that element filters slow physiological signals as well as a conventional filter, while banks of mixed compositions add little beyond linear filtering at the same time constants. Filtering a recorded signal with a fabricated device, measuring its small-signal relaxation, and comparing it with a printed RC element would test the practical value of the chemical control.

References

1. Prado-Socorro, C. D., Giménez-Santamarina, S., Mardegan, L., *et al.* Polymer-Based Composites for Engineering Organic Memristive Devices. *Advanced Electronic Materials* **8,** 2101192. doi:10.1002/aelm.202101192 (2022).

2. Roy, K., Jaiswal, A. & Panda, P. Towards spike-based machine intelligence with neuromorphic computing. *Nature* **575,** 607–617. doi:10.1038/s41586-019-1677-2 (2019).

3. Indiveri, G. & Liu, S.-C. Memory and Information Processing in Neuromorphic Systems. *Proceedings of the IEEE* **103,** 1379–1397. doi:10.1109/JPROC.2015.2444094 (2015).

4. Tanaka, G., Yamane, T., Héroux, J. B., *et al.* Recent advances in physical reservoir computing: A review. *Neural Networks* **115,** 100–123. doi:10.1016/j.neunet.2019.03.005 (2019).

5. Liang, X., Tang, J., Zhong, Y., Gao, B., Qian, H. & Wu, H. Physical reservoir computing with emerging electronics. *Nature Electronics* **7,** 193–206. doi:10.1038/s41928-024-01133-z (2024).

6. Indiveri, G. & Sandamirskaya, Y. The Importance of Space and Time for Signal Processing in Neuromorphic Agents: The Challenge of Developing Low-Power, Autonomous Agents That Interact With the Environment. *IEEE Signal Processing Magazine* **36,** 16–28. doi:10.1109/MSP.2019.2928376 (2019).

7. Zhao, H., Scholz, A., Beigl, M., Ni, S., Singaraju, S. A. & Aghassi-Hagmann, J. *Printed Electrodermal Activity Sensor with Optimized Filter for Stress Detection* in *Proceedings of the 2022 ACM International Symposium on Wearable Computers (ISWC '22)* (ACM, 2022), 112–114. doi:10.1145/3544794.3558479.

8. Zhao, H., Pal, P., Hefenbrock, M., Beigl, M. & Tahoori, M. B. *Towards Temporal Information Processing – Printed Neuromorphic Circuits with Learnable Filters* in *Proceedings of the 18th ACM International Symposium on Nanoscale Architectures (NANOARCH '23)* (ACM, 2023), 1–6. doi:10.1145/3611315.3633249.

9. Du, C., Cai, F., Zidan, M. A., Ma, W., Lee, S. H. & Lu, W. D. Reservoir computing using dynamic memristors for temporal information processing. *Nature Communications* **8,** 2204. doi:10.1038/s41467-017-02337-y (2017).

10. Zhong, Y., Tang, J., Li, X., Gao, B., Qian, H. & Wu, H. Dynamic memristor-based reservoir computing for high-efficiency temporal signal processing. *Nature Communications* **12,** 408. doi:10.1038/s41467-020-20692-1 (2021).

11. Cucchi, M., Gruener, C., Petrauskas, L., *et al.* Reservoir computing with biocompatible organic electrochemical networks for brain-inspired bio-signal classification. *Science Advances* **7,** eabh0693. doi:10.1126/sciadv.abh0693 (2021).

12. Appeltant, L., Soriano, M. C., Van der Sande, G., *et al.* Information processing using a single dynamical node as complex system. *Nature Communications* **2,** 468. doi:10.1038/ncomms1476 (2011).

13. Lee, S. H., Kim, S., Han, J., *et al.* Electrically tunable physical reservoir computing on two-terminal devices. *npj Unconventional Computing* **2,** 27. doi:10.1038/s44335-025-00043-3 (2025).

14. Gao, C., Liu, D., Xu, C., *et al.* Toward grouped-reservoir computing: organic neuromorphic vertical transistor with distributed reservoir states for efficient recognition and prediction. *Nature Communications* **15,** 740. doi:10.1038/s41467-024-44942-8 (2024).

15. Han, C., Kim, M., Park, S. J., *et al.* Reconfigurable Organic Electrochemical Transistors with High Dynamic Ranges for Fully Integrated Physical Reservoir Computing. *Advanced Functional Materials* **35,** 2423814. doi:10.1002/adfm.202423814 (2025).

16. Maass, W., Natschläger, T. & Markram, H. Real-Time Computing Without Stable States: A New Framework for Neural Computation Based on Perturbations. *Neural Computation* **14,** 2531–2560. doi:10.1162/089976602760407955 (2002).

17. Van de Burgt, Y., Lubberman, E., Fuller, E. J., *et al.* A non-volatile organic electrochemical device as a low-voltage artificial synapse for neuromorphic computing. *Nature Materials* **16,** 414–418. doi:10.1038/nmat4856 (2017).

18. Van de Burgt, Y., Melianas, A., Keene, S. T., Malliaras, G. & Salleo, A. Organic electronics for neuromorphic computing. *Nature Electronics* **1,** 386–397. doi:10.1038/s41928-018-0103-3 (2018).

19. Fuller, E. J., Keene, S. T., Melianas, A., *et al.* Parallel programming of an ionic floating-gate memory array for scalable neuromorphic computing. *Science* **364,** 570–574. doi:10.1126/science.aaw5581 (2019).

20. Gkoupidenis, P., Koutsouras, D. A. & Malliaras, G. G. Neuromorphic device architectures with global connectivity through electrolyte gating. *Nature Communications* **8,** 15448. doi:10.1038/ncomms15448 (2017).